# SNx4AHC02 Quadruple 2-Input Positive-Nor Gates

## 1 Features

- Operating range 2V to 5.5V
- Latch-up performance exceeds 250mA per JESD 17

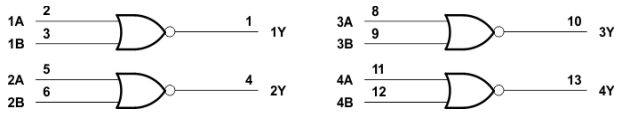
## 2 Description

The ’AHC02 devices contain four independent 2-input NOR gates that perform the Boolean function $Y = \overline{A} \bullet \overline{B}$ or $Y = \overline{A + B}$ in positive logic.

**Device Information**

| PART NUMBER | PACKAGE(1) | PACKAGE SIZE(2) | BODY SIZE(3) |
|---|---|---|---|
| SNx4AHC02 | D (SOIC, 14) | 8.65mm × 6mm | 8.65mm × 3.9mm |
| | DB (SSOP, 14) | 6.20mm × 7.8mm | 6.20mm × 5.3mm |
| | DGV (TVSOP, 14) | 3.60mm × 6.4mm | 3.60mm × 4.4mm |
| | N (PDIP, 14) | 19.30mm × 9.4mm | 19.30mm × 6.35mm |
| | NS (SOP, 14) | 10.2mm × 7.8mm | 10.30mm ×5.3mm |
| | PW (SOP, 14) | 5.00mm × 6.4mm | 5.00mm × 4.4mm |
| | RGY (VQFN, 14) | 3.50mm × 3.5mm | 3.50mm × 3.5mm |
| | BQA (WQFN, 14) | 3mm × 2.5mm | 3mm × 2.5mm |

(1) For more information, see Section 10.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.
(3) The body size (length × width) is a nominal value and does not include pins.




Pin numbers shown are for the D, DB, DGV, J, N, NS, PW, RGY, and W packages.

**Logic Diagram (Positive Logic)**

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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## 3 Pin Configuration and Functions

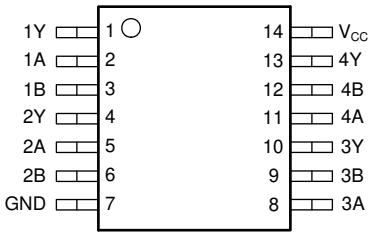



**Figure 3-1. SN54AHC02 J or W Package, 14-Pin (Top View)**
**SN74AHC02 D, DB, DGV, N, NS, or PW Package, 14-Pin (Top View)**

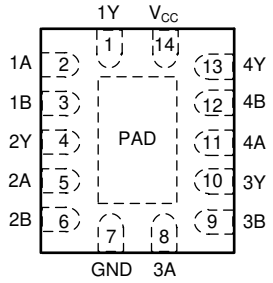



**Figure 3-2. SN74AHC02 RGY or BQA Package, 14-Pin (Top View)**

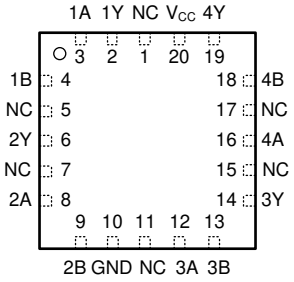



**Figure 3-3. SN54AHC02 FK Package, 20-Pin (Top View)**

**Table 3-1. Pin Functions**

| PIN | | | | TYPE[1] | DESCRIPTION |
|---|---|---|---|---|---|
| NAME | SN74AHC02 | SN54AHC02 | | | |
| | D, DB, DGV, N, NS, PW, RGY, BQA | J, W | FK | | |
| 1A | 2 | 2 | 3 | I | 1A Input |
| 1B | 3 | 3 | 4 | I | 1B Input |
| 1Y | 1 | 1 | 2 | O | 1Y Output |
| 2A | 5 | 5 | 8 | I | 2A Input |
| 2B | 6 | 6 | 9 | I | 2B Input |
| 2Y | 4 | 4 | 6 | O | 2Y Output |
| 3A | 8 | 8 | 12 | I | 3A Input |
| 3B | 9 | 9 | 13 | I | 3B Input |
| 3Y | 10 | 20 | 14 | O | 3Y Output |
| 4A | 11 | 11 | 16 | I | 4A Input |
| 4B | 12 | 12 | 18 | I | 4B Input |
| 4Y | 13 | 13 | 19 | O | 4Y Output |
| GND | 7 | 7 | 10 | — | Ground Pin |
| NC | — | — | 1, 5, 7, 11, 15, 17 | — | No Connection |
| $V_{CC}$ | 14 | 14 | 20 | — | Power Pin |

(1) I = input, O = output

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# 4 Specifications

## 4.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

| | | | MIN | MAX | UNIT |
|---|---|---|---|---|---|
| $V_{CC}$ | Supply voltage range | | –0.5 | 7 | V |
| $V_I$ (2) | Input voltage range | | –0.5 | 7 | V |
| $V_O$ (2) | Output voltage range | | –0.5 | $V_{CC}$ + 0.5 | V |
| $I_{IK}$ | Input clamp current | ($V_I$ < 0) | | –20 | mA |
| $I_{OK}$ | Output clamp current | ($V_O$ < 0 or $V_O$ > $V_{CC}$) | | ±20 | mA |
| $I_O$ | Continuous output current | ($V_O$ = 0 to $V_{CC}$) | | ±25 | mA |
| | Continuous current through $V_{CC}$ or GND | | | ±50 | mA |
| $T_{stg}$ | Storage temperature range | | –65 | 150 | °C |

(1) Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

## 4.2 ESD Ratings

| | | | VALUE | UNIT |
|---|---|---|---|---|
| $V_{(ESD)}$ | Electrostatic discharge | Human-body model (HBM)(1) | ±2000 | V |
| | | Charged device model (CDM)(2) | ±1000 | |

(1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

## 4.3 Recommended Operating Conditions

| | | | SN54AHC02 | | SN74AHC02 | | UNIT |
|---|---|---|---|---|---|---|---|
| | | | MIN | MAX | MIN | MAX | |
| $V_{CC}$ | Supply voltage | | 2 | 5.5 | 2 | 5.5 | V |
| $V_{IH}$ | High-level input voltage | $V_{CC}$= 2 V | 1.5 | | 1.5 | | V |
| | | $V_{CC}$= 3V | 2.1 | | 2.1 | | |
| | | $V_{CC}$= 5.5 V | 3.85 | | 3.85 | | |
| $V_{IL}$ | Low-level Input voltage | $V_{CC}$= 2 V | | 0.5 | | 0.5 | V |
| | | $V_{CC}$= 3 V | | 0.9 | | 0.9 | |
| | | $V_{CC}$= 5.5 V | | 1.65 | | 1.65 | |
| $V_I$ | Input voltage | | 0 | 5.5 | 0 | 5.5 | V |
| $V_O$ | Output voltage | | 0 | $V_{CC}$ | 0 | $V_{CC}$ | V |
| $I_{OH}$ | High-level output current | $V_{CC}$= 2 V | | –50 | | –50 | mA |
| | | $V_{CC}$= 3.3 V ± 0.3 V | | –4 | | –4 | |
| | | $V_{CC}$= 5 V ± 0.5 V | | –8 | | –8 | |
| $I_{OL}$ | Low-level output current | $V_{CC}$= 2 V | | 50 | | 50 | mA |
| | | $V_{CC}$= 3.3 V ± 0.3 V | | 4 | | 4 | |
| | | $V_{CC}$= 5 V ± 0.5 V | | 8 | | 8 | |
| Δt/Δv | Input Transition rise or fall rate | $V_{CC}$= 3.3 V ± 0.3 V | | 100 | | 100 | ns/V |
| | | $V_{CC}$= 5 V ± 0.5 V | | 20 | | 20 | |
| $T_A$ | Operating free-air temperature | | –55 | 125 | –40 | 125 | °C |


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## 4.4 Thermal Information

| THERMAL METRIC[1] | | SN74AHC02 | | | | | | | | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|
| | | D | DB | DGV | N | NS | PW | RGY | BQA | |
| | | 14 PINS | 14 PINS | 14 PINS | 14 PINS | 14 PINS | 14 PINS | 14 PINS | 14 PINS | |
| $R_{\theta JA}$ | Junction-to-ambient thermal resistance | 124.6 | 96 | 127 | 80 | 76 | 147.7 | 47 | 88.3 | °C/W |

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, SPRA953.

## 4.5 Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)[1]

| PARAMETER | TEST CONDITIONS | $V_{CC}$ | $T_A$ = 25°C | | | $T_A$ = –55°C TO 125°C | | $T_A$ = –40°C TO 85°C | | $T_A$ = –40°C TO 125°C | | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | SN54AHC02 | | SN74AHC02 | | Recommended SN74AHC02 | | |
| | | | MIN | TYP | MAX | MIN | MAX | MIN | MAX | MIN | MAX | |
| $V_{OH}$ | $I_{OH}$ = –50 µA | 2 V | 1.9 | 2 | | 1.9 | | 1.9 | | 1.9 | | V |
| | | 3 V | 2.9 | 3 | | 2.9 | | 2.9 | | 2.9 | | |
| | | 4.5 V | 4.4 | 4.5 | | 4.4 | | 4.4 | | 4.4 | | |
| | $I_{OH}$ = –4 mA | 3 V | 2.58 | | | 2.48 | | 2.48 | | 2.48 | | |
| | $I_{OH}$ = –8 mA | 4.5 V | 3.94 | | | 3.8 | | 3.8 | | 3.8 | | |
| $V_{OL}$ | $I_{OL}$ = 50 µA | 2 V | | | 0.1 | | 0.1 | | 0.1 | | 0.1 | V |
| | | 3 V | | | 0.1 | | 0.1 | | 0.1 | | 0.1 | |
| | | 4.5 V | | | 0.1 | | 0.1 | | 0.1 | | 0.1 | |
| | $I_{OH}$ = 4 mA | 3 V | | | 0.36 | | 0.5 | | 0.44 | | 0.5 | |
| | $I_{OH}$ = 8 mA | 4.5 V | | | 0.36 | | 0.5 | | 0.44 | | 0.5 | |
| $I_I$ | $V_I$ = 5.5 V or GND | 0 V to 5.5 V | | | ±0.1 | | ±1[1] | | ±1 | | ±1 | µA |
| $I_{CC}$ | $V_I$ = $V_{CC}$ or GND, $I_O$ = 0 | 5.5 V | | | 2 | | 20 | | 20 | | 20 | µA |
| $C_i$ | $V_I$ = $V_{CC}$ or GND | 5 V | | 4 | 10 | | | | 10 | | | pF |

(1) On products compliant to MIL-PRF-38535, this parameter is not production tested at VCC = 0 V.

## 4.6 Switching Characteristics, $V_{CC}$ = 3.3 V ± 0.3 V

over recommended operating free-air temperature range, $V_{CC}$ = 3.3 V ± 0.3 V (unless otherwise noted) (see Load Circuit and Voltage Waveforms)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | LOAD CAPACITANCE | $T_A$ = 25°C | | $T_A$ = –55°C TO 125°C | | $T_A$ = –40°C TO 85°C | | $T_A$ = –40°C TO 125°C | | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | SN54AHC02 | | SN74AHC02 | | Recommended SN74AHC02 | | |
| | | | | TYP | MAX | MIN | MAX | MIN | MAX | MIN | MAX | |
| $t_{PLH}$ | A or B | Y | $C_L$ = 15 pF | 5.6[1] | 7.9[1] | 1[1] | 9.5[1] | 1 | 9.5 | 1 | 9.5 | ns |
| $t_{PHL}$ | | | | 5.6[1] | 7.9[1] | 1[1] | 9.5[1] | 1 | 9.5 | 1 | 9.5 | |
| $t_{PLH}$ | A or B | Y | $C_L$ = 50 pF | 8.1 | 11.4 | 1 | 13 | 1 | 13 | 1 | 13 | ns |
| $t_{PHL}$ | | | | 8.1 | 11.4 | 1 | 13 | 1 | 13 | 1 | 13 | |

(1) On products compliant to MIL-PRF-38535, this parameter is not production tested.

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## 4.7 Switching Characteristics, $V_{CC}$ = 5 V ± 0.5 V

over recommended operating free-air temperature range, $V_{CC}$ = 5 V ± 0.5 V (unless otherwise noted) (see Load Circuit and Voltage Waveforms)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | LOAD CAPACITANCE | $T_A$ = 25°C | | $T_A$ = –55°C TO 125°C | | $T_A$ = –40°C TO 85°C | | $T_A$ = –40°C TO 125°C | | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | SN54AHC02 | | SN74AHC02 | | Recommended SN74AHC02 | | |
| | | | | TYP | MAX | MIN | MAX | MIN | MAX | MIN | MAX | |
| $t_{PLH}$ | A or B | Y | $C_L$ = 15 pF | 3.6[1] | 5.5[1] | 1[1] | 6.5[1] | 1 | 6.5 | 1 | 6.5 | ns |
| $t_{PHL}$ | | | | 3.6[1] | 5.5[1] | 1[1] | 6.5[1] | 1 | 6.5 | 1 | 6.5 | |
| $t_{PLH}$ | A or B | Y | $C_L$ = 50 pF | 5.1 | | 1 | 8.5 | 1 | 8.5 | 1 | 8.5 | ns |
| $t_{PHL}$ | | | | 5.1 | | 1 | 8.5 | 1 | 8.5 | 1 | 8.5 | |

## 4.8 Noise Characteristics

$V_{CC}$ = 5 V, $C_L$ = 50 pF, $T_A$ = 25°C[1]

| PARAMETER | | SN74AHC02 | | UNIT |
|---|---|---|---|---|
| | | MIN | MAX | |
| $V_{OL(P)}$ | Quiet output, maximum dynamic $V_{OL}$ | | 0.8 | V |
| $V_{OL(V)}$ | Quiet output, minimum dynamic $V_{OL}$ | | –0.8 | V |
| $V_{OH(V)}$ | Quiet output, minimum dynamic $V_{OH}$ | 4.9 | | V |
| $V_{IH(D)}$ | High-level dynamic input voltage | 3.5 | | V |
| $V_{IL(D)}$ | Low-level dynamic input voltage | | 1.5 | V |

(1) Characteristics are for surface-mount packages only.

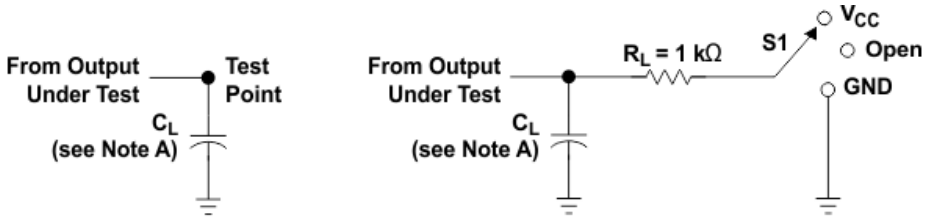
## 4.9 Operating Characteristics

$V_{CC}$ = 5 V, $T_A$ = 25°C

| PARAMETER | | TEST CONDITIONS | | TYP | UNIT |
|---|---|---|---|---|---|
| $C_{pd}$ | Power dissipation capacitance | No load, | f = 1 MHz | 15 | pF |


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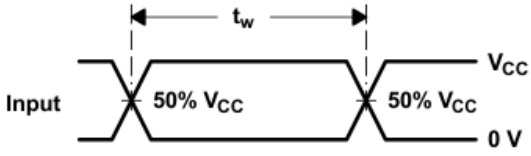
## 5 Parameter Measurement Information



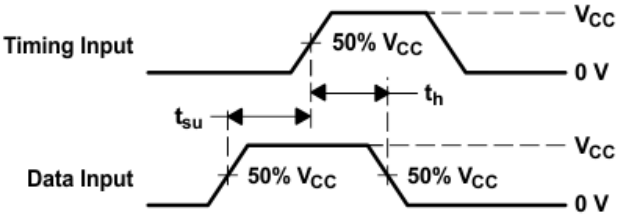

| TEST | S1 |
|---|---|
| $t_{PLH}/t_{PHL}$ | Open |
| $t_{PLZ}/t_{PZL}$ | $V_{CC}$ |
| $t_{PHZ}/t_{PZH}$ | GND |
| Open Drain | $V_{CC}$ |

LOAD CIRCUIT FOR TOTEM-POLE OUTPUTS

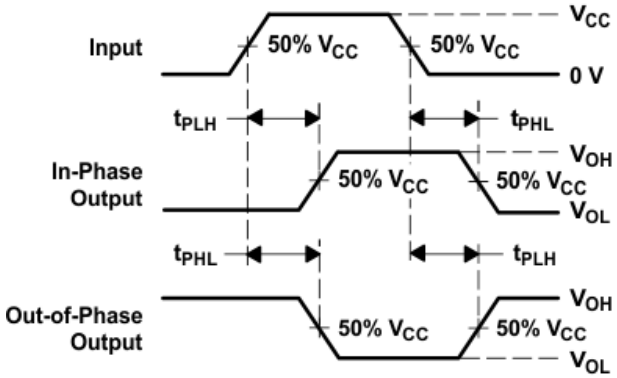
LOAD CIRCUIT FOR 3-STATE AND OPEN-DRAIN OUTPUTS



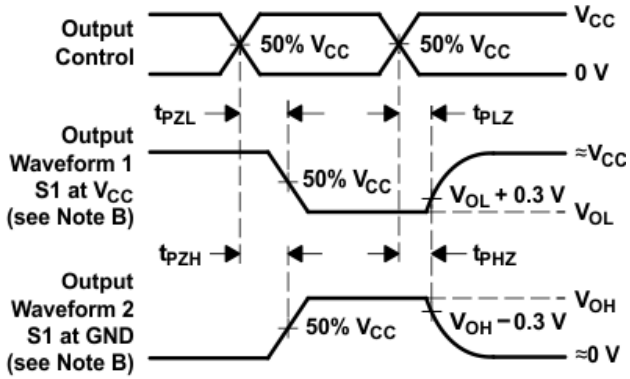

VOLTAGE WAVEFORMS PULSE DURATION




VOLTAGE WAVEFORMS SETUP AND HOLD TIMES




VOLTAGE WAVEFORMS PROPAGATION DELAY TIMES INVERTING AND NONINVERTING OUTPUTS




VOLTAGE WAVEFORMS ENABLE AND DISABLE TIMES LOW- AND HIGH-LEVEL ENABLING

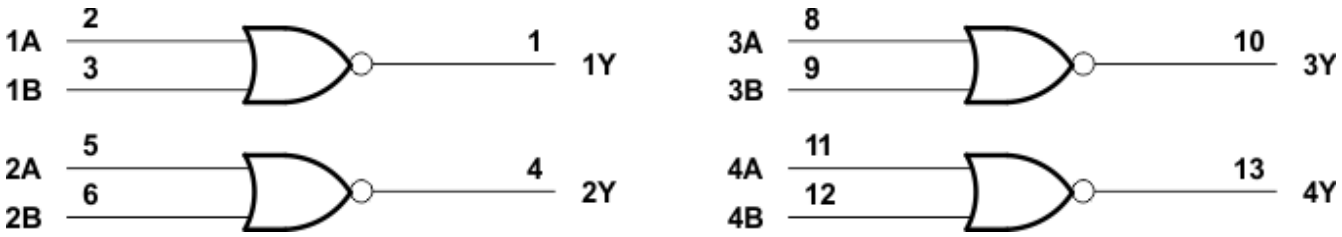
A. $C_L$ includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
C. All input pulses are supplied by generators having the following characteristics: PRR ≤ 1 MHz, $Z_O$ = 50 Ω, $t_r$ ≤ 3 ns, $t_f$ ≤ 3 ns.
D. The outputs are measured one at a time with one input transition per measurement.
E. All parameters and waveforms are not applicable to all devices.

**Figure 5-1. Load Circuit and Voltage Waveforms**

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# 6 Detailed Description

## 6.1 Functional Block Diagram




## 6.2 Device Functional Modes

**Table 6-1. Function Table**

| INPUTS[1] | | OUTPUT Y |
|---|---|---|
| A | B | |
| H | X | L |
| X | H | L |
| L | L | H |

(1) H = High Voltage Level, L = Low Voltage Level, X = Don't Care


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# 7 Application and Implementation

**Note**

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

## 7.1 Power Supply Recommendations

The power supply can be any voltage between the MIN and MAX supply-voltage rating located in *Section 4.3*.

The power supply can be any voltage between the minimum and maximum supply voltage rating located in the *Recommended Operating Conditions*. Each $V_{CC}$ terminal should have a good bypass capacitor to prevent power disturbance. A 0.1-µF capacitor is recommended for this device. It is acceptable to parallel multiple bypass capacitors to reject different frequencies of noise. The 0.1-µF and 1-µF capacitors are commonly used in parallel. The bypass capacitor should be installed as close to the power terminal as possible for best results, as shown in *Layout Example*.

## 7.2 Layout

### 7.2.1 Layout Guidelines

When using multiple-input and multiple-channel logic devices, inputs must never be left floating. In many cases, functions or parts of functions of digital logic devices are unused (for example, when only two inputs of a triple-input AND gate are used or only 3 of the 4 buffer gates are used). Such unused input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. All unused inputs of digital logic devices must be connected to a logic high or logic low voltage, as defined by the input voltage specifications, to prevent them from floating. The logic level that must be applied to any particular unused input depends on the function of the device. Generally, the inputs are tied to GND or $V_{CC}$, whichever makes more sense for the logic function or is more convenient.

### 7.2.2 Layout Example

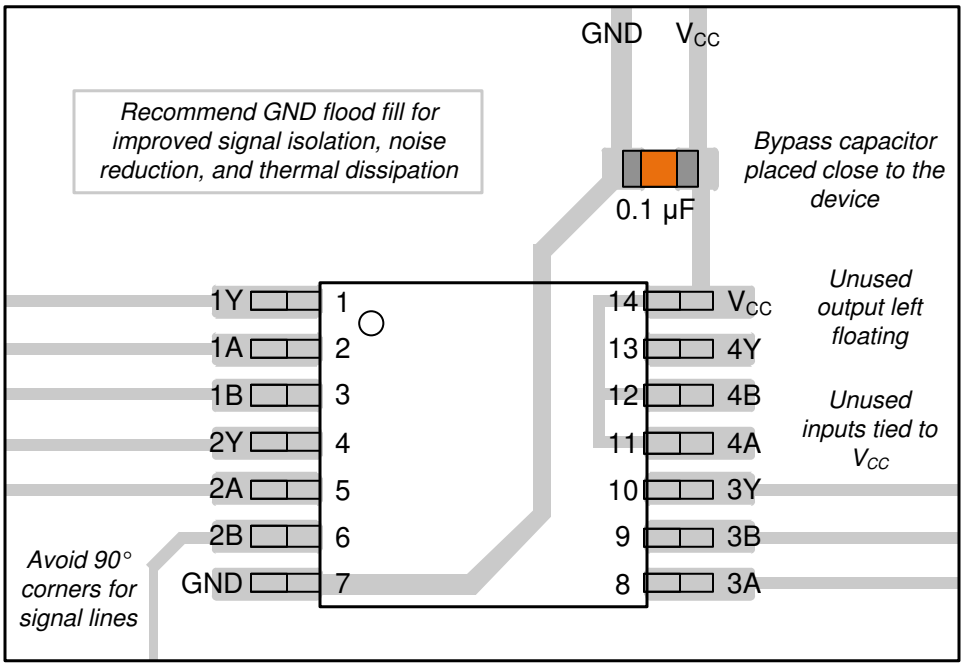



**Figure 7-1. Example Layout for the SN74AHC02**

# 8 Device and Documentation Support

TI offers an extensive line of development tools. Tools and software to evaluate the performance of the device, generate code, and develop solutions are listed below.

## 8.1 Documentation Support

### 8.1.1 Related Documentation

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

**Table 8-1. Related Links**

| PARTS | PRODUCT FOLDER | SAMPLE & BUY | TECHNICAL DOCUMENTS | TOOLS & SOFTWARE | SUPPORT & COMMUNITY |
|---|---|---|---|---|---|
| SN54AHC02 | Click here | Click here | Click here | Click here | Click here |
| SN74AHC02 | Click here | Click here | Click here | Click here | Click here |

## 8.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

## 8.3 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

## 8.4 Trademarks

TI E2E™ is a trademark of Texas Instruments.
All trademarks are the property of their respective owners.

## 8.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

## 8.6 Glossary

TI Glossary — This glossary lists and explains terms, acronyms, and definitions.

# 9 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Changes from Revision M (June 2023) to Revision N (February 2024) | Page |
|---|---|
| • Added package size to *Device Information* table | 1 |
| • Updated RθJA value: D = 124.5 to 124.6, all values in °C/W | 5 |
| • Added *Application and Implementation* section | 9 |


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| Changes from Revision L (May 2013) to Revision M (June 2023) | Page |
|---|---|
| • Added *Package Information* table, *Pin Functions* table, *ESD Ratings* table, *Thermal Information* table, *Device Functional Modes, Device and Documentation Support* section, and *Mechanical, Packaging, and Orderable Information* section | 1 |
| • Added BQA package to *Package Information* table | 1 |
| • Updated thermal values for RθJA: D = 86 to 124.5, PW = 113 to 147.7, all values in °C/W | 5 |
| • Added thermal value for RθJA: BQA = 88.3, all values in °C/W | 5 |

## 10 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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## PACKAGING INFORMATION

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| 5962-9752801Q2A | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801Q2A SNJ54AHC02FK |
| 5962-9752801QCA | Active | Production | CDIP (J) \| 14 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801QCA SNJ54AHC02J |
| 5962-9752801QDA | Active | Production | CFP (W) \| 14 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801QDA SNJ54AHC02W |
| SN74AHC02BQAR | Active | Production | WQFN (BQA) \| 14 | 3000 \| LARGE T&R | Yes | NIPDAU \| SN | Level-1-260C-UNLIM | -40 to 125 | AHC02 |
| SN74AHC02BQAR.A | Active | Production | WQFN (BQA) \| 14 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | AHC02 |
| SN74AHC02D | Obsolete | Production | SOIC (D) \| 14 | - | - | Call TI | Call TI | -40 to 125 | AHC02 |
| SN74AHC02DBR | Active | Production | SSOP (DB) \| 14 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | HA02 |
| SN74AHC02DBR.A | Active | Production | SSOP (DB) \| 14 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | HA02 |
| SN74AHC02DGVR | Active | Production | TVSOP (DGV) \| 14 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | HA02 |
| SN74AHC02DGVR.A | Active | Production | TVSOP (DGV) \| 14 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | HA02 |
| SN74AHC02DR | Active | Production | SOIC (D) \| 14 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | AHC02 |
| SN74AHC02DR.A | Active | Production | SOIC (D) \| 14 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | AHC02 |
| SN74AHC02N | Active | Production | PDIP (N) \| 14 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -40 to 125 | SN74AHC02N |
| SN74AHC02N.A | Active | Production | PDIP (N) \| 14 | 25 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | -40 to 125 | SN74AHC02N |
| SN74AHC02NSR | NRND | Production | SOP (NS) \| 14 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | AHC02 |
| SN74AHC02NSR.A | NRND | Production | SOP (NS) \| 14 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | AHC02 |
| SN74AHC02PW | Obsolete | Production | TSSOP (PW) \| 14 | - | - | Call TI | Call TI | -40 to 125 | HA02 |
| SN74AHC02PWR | Active | Production | TSSOP (PW) \| 14 | 2000 \| LARGE T&R | Yes | NIPDAU \| SN | Level-1-260C-UNLIM | -40 to 125 | HA02 |
| SN74AHC02PWR.A | Active | Production | TSSOP (PW) \| 14 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | HA02 |
| SN74AHC02RGYR | Last Time Buy | Production | VQFN (RGY) \| 14 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-2-260C-1 YEAR | -40 to 125 | HA02 |
| SN74AHC02RGYR.A | Last Time Buy | Production | VQFN (RGY) \| 14 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-2-260C-1 YEAR | -40 to 125 | HA02 |

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| SNJ54AHC02FK | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801Q2A SNJ54AHC 02FK |
| SNJ54AHC02FK.A | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801Q2A SNJ54AHC 02FK |
| SNJ54AHC02J | Active | Production | CDIP (J) \| 14 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801QC A SNJ54AHC02J |
| SNJ54AHC02J.A | Active | Production | CDIP (J) \| 14 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801QC A SNJ54AHC02J |
| SNJ54AHC02W | Active | Production | CFP (W) \| 14 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801QD A SNJ54AHC02W |
| SNJ54AHC02W.A | Active | Production | CFP (W) \| 14 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9752801QD A SNJ54AHC02W |

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

**Important Information and Disclaimer:**The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

**OTHER QUALIFIED VERSIONS OF SN54AHC02, SN74AHC02 :**

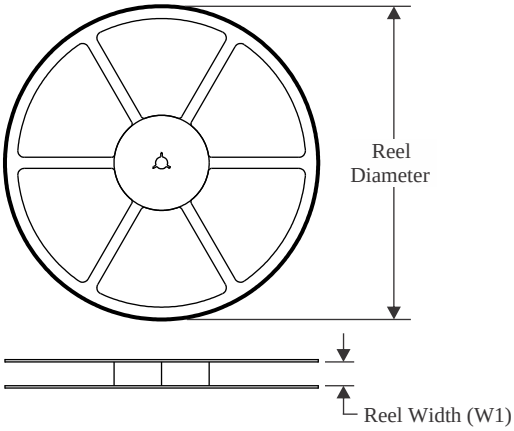
- Catalog : SN74AHC02
- Automotive : SN74AHC02-Q1, SN74AHC02-Q1
- Enhanced Product : SN74AHC02-EP, SN74AHC02-EP
- Military : SN54AHC02

NOTE: Qualified Version Definitions:

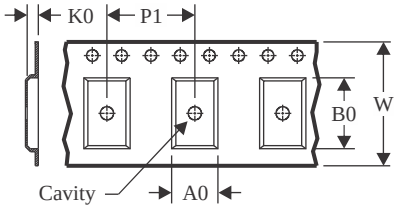
- Catalog - TI's standard catalog product
- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects
- Enhanced Product - Supports Defense, Aerospace and Medical Applications
- Military - QML certified for Military and Defense Applications

## TAPE AND REEL INFORMATION

**REEL DIMENSIONS**

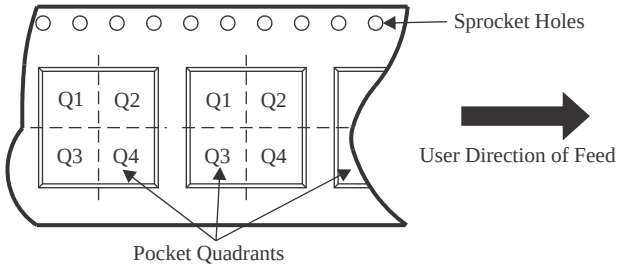



**TAPE DIMENSIONS**




| | |
|---|---|
| A0 | Dimension designed to accommodate the component width |
| B0 | Dimension designed to accommodate the component length |
| K0 | Dimension designed to accommodate the component thickness |
| W | Overall width of the carrier tape |
| P1 | Pitch between successive cavity centers |

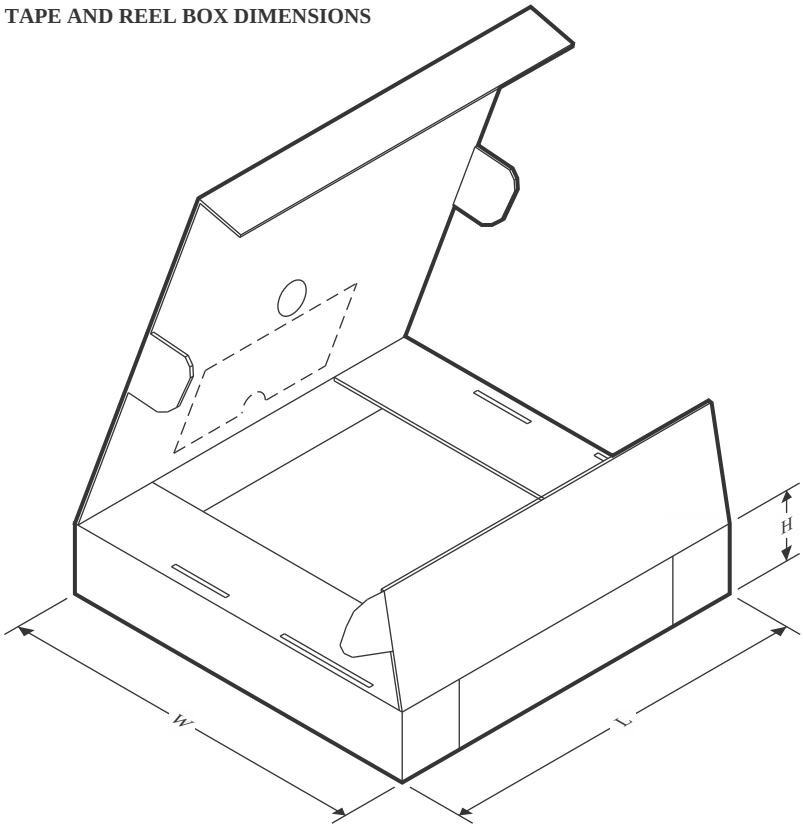
**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SN74AHC02BQAR | WQFN | BQA | 14 | 3000 | 177.8 | 12.4 | 2.8 | 3.3 | 1.1 | 4.0 | 12.0 | Q1 |
| SN74AHC02DBR | SSOP | DB | 14 | 2000 | 330.0 | 16.4 | 8.35 | 6.6 | 2.4 | 12.0 | 16.0 | Q1 |
| SN74AHC02DGVR | TVSOP | DGV | 14 | 2000 | 330.0 | 12.4 | 6.8 | 4.0 | 1.6 | 8.0 | 12.0 | Q1 |
| SN74AHC02NSR | SOP | NS | 14 | 2000 | 330.0 | 16.4 | 8.1 | 10.4 | 2.5 | 12.0 | 16.0 | Q1 |
| SN74AHC02PWR | TSSOP | PW | 14 | 2000 | 330.0 | 12.4 | 6.9 | 5.6 | 1.6 | 8.0 | 12.0 | Q1 |
| SN74AHC02RGYR | VQFN | RGY | 14 | 3000 | 330.0 | 12.4 | 3.75 | 3.75 | 1.15 | 8.0 | 12.0 | Q1 |

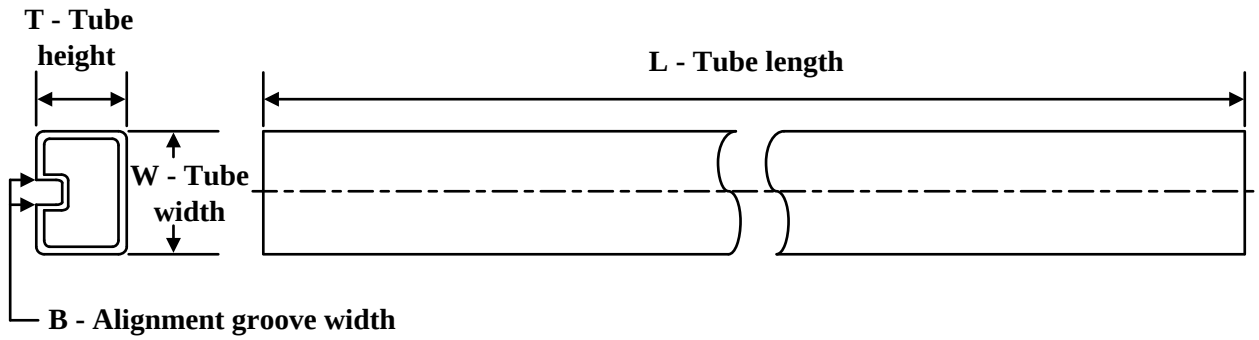
TAPE AND REEL BOX DIMENSIONS




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| SN74AHC02BQAR | WQFN | BQA | 14 | 3000 | 208.0 | 191.0 | 35.0 |
| SN74AHC02DBR | SSOP | DB | 14 | 2000 | 353.0 | 353.0 | 32.0 |
| SN74AHC02DGVR | TVSOP | DGV | 14 | 2000 | 353.0 | 353.0 | 32.0 |
| SN74AHC02NSR | SOP | NS | 14 | 2000 | 353.0 | 353.0 | 32.0 |
| SN74AHC02PWR | TSSOP | PW | 14 | 2000 | 353.0 | 353.0 | 32.0 |
| SN74AHC02RGYR | VQFN | RGY | 14 | 3000 | 353.0 | 353.0 | 32.0 |

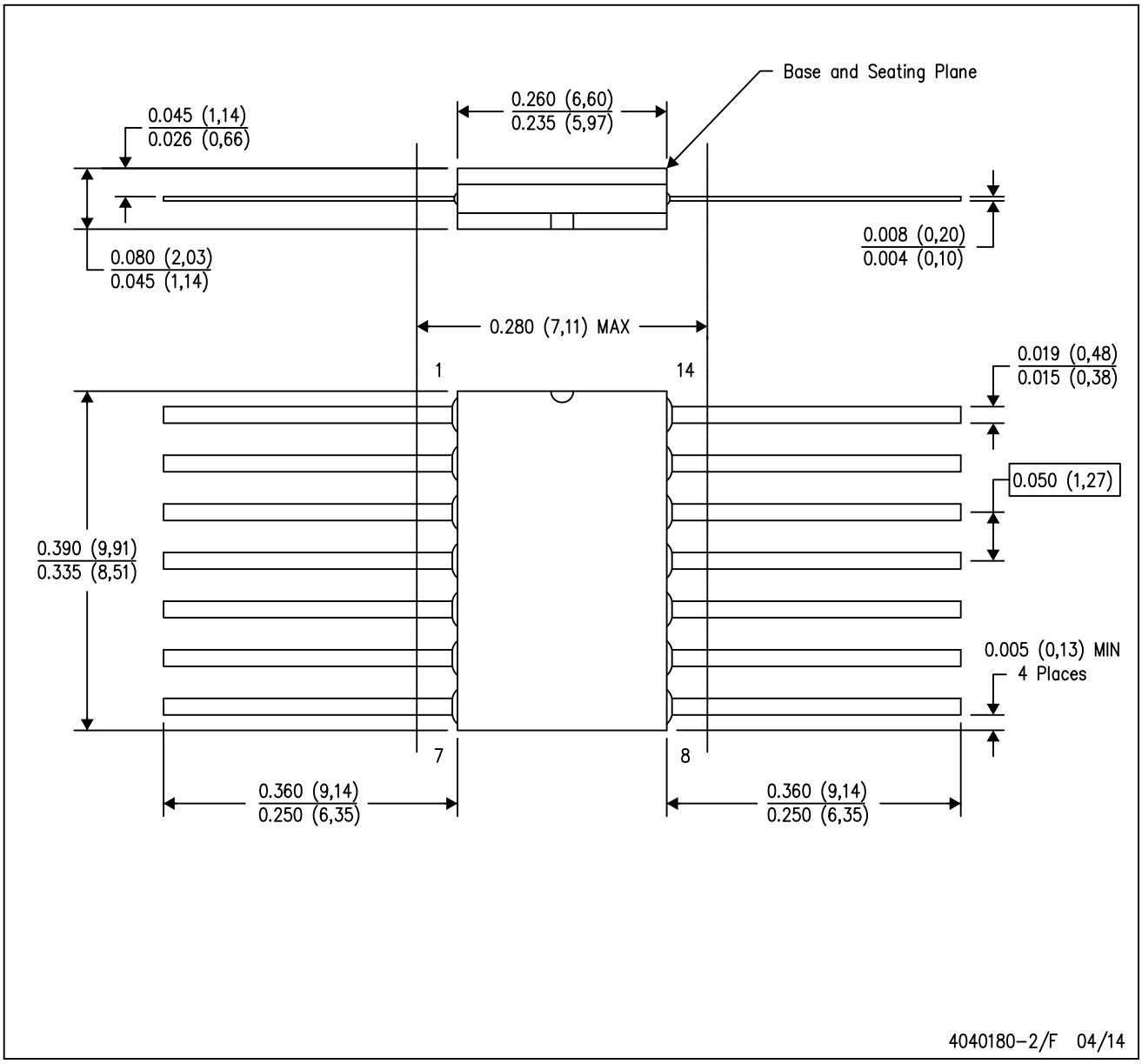
## TUBE




*All dimensions are nominal

| Device | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (µm) | B (mm) |
|---|---|---|---|---|---|---|---|---|
| 5962-9752801Q2A | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| 5962-9752801QDA | W | CFP | 14 | 25 | 506.98 | 26.16 | 6220 | NA |
| SN74AHC02N | N | PDIP | 14 | 25 | 506 | 13.97 | 11230 | 4.32 |
| SN74AHC02N | N | PDIP | 14 | 25 | 506 | 13.97 | 11230 | 4.32 |
| SN74AHC02N.A | N | PDIP | 14 | 25 | 506 | 13.97 | 11230 | 4.32 |
| SN74AHC02N.A | N | PDIP | 14 | 25 | 506 | 13.97 | 11230 | 4.32 |
| SNJ54AHC02FK | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| SNJ54AHC02FK.A | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| SNJ54AHC02W | W | CFP | 14 | 25 | 506.98 | 26.16 | 6220 | NA |
| SNJ54AHC02W.A | W | CFP | 14 | 25 | 506.98 | 26.16 | 6220 | NA |

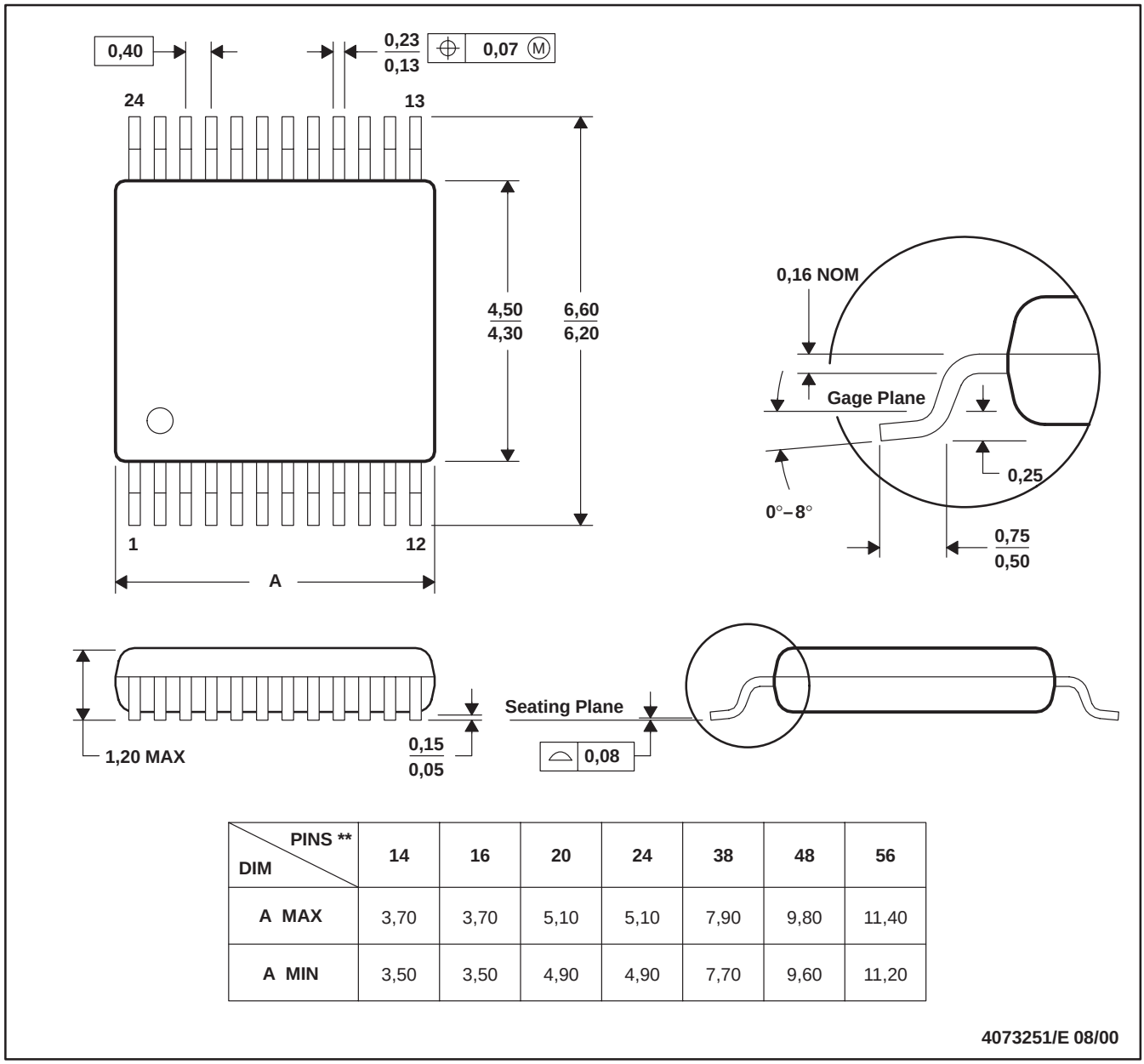
W (R–GDFP–F14) CERAMIC DUAL FLATPACK




NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. This package can be hermetically sealed with a ceramic lid using glass frit.
D. Index point is provided on cap for terminal identification only.
E. Falls within MIL STD 1835 GDFP1–F14

**DGV (R-PDSO-G**)** **PLASTIC SMALL-OUTLINE**

**24 PINS SHOWN**




| DIM \ PINS ** | 14 | 16 | 20 | 24 | 38 | 48 | 56 |
|---|---|---|---|---|---|---|---|
| A MAX | 3,70 | 3,70 | 5,10 | 5,10 | 7,90 | 9,80 | 11,40 |
| A MIN | 3,50 | 3,50 | 4,90 | 4,90 | 7,70 | 9,60 | 11,20 |

**4073251/E 08/00**

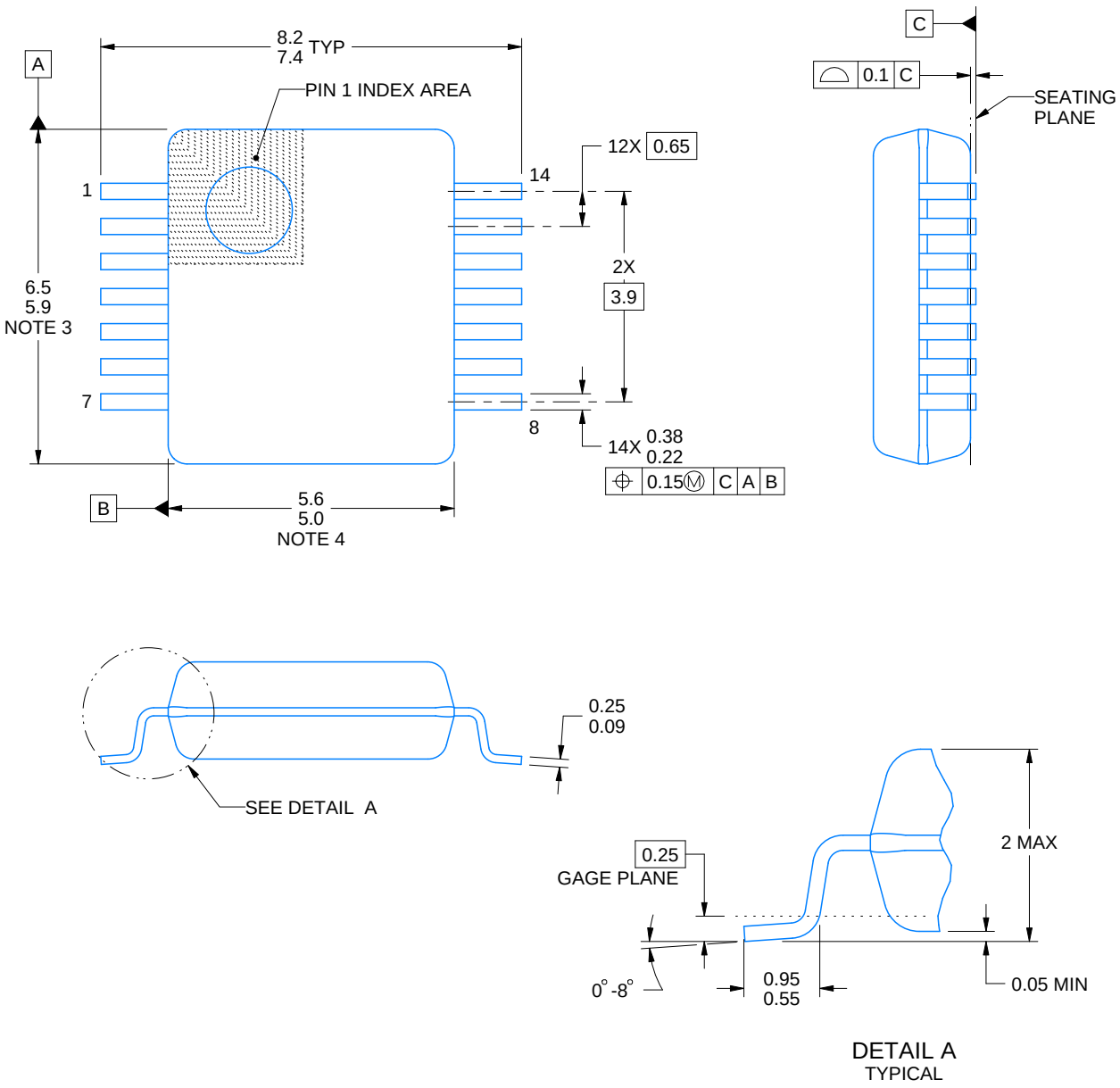
NOTES: A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15 per side.
D. Falls within JEDEC: 24/48 Pins – MO-153
14/16/20/56 Pins – MO-194

# PACKAGE OUTLINE

## DB0014A

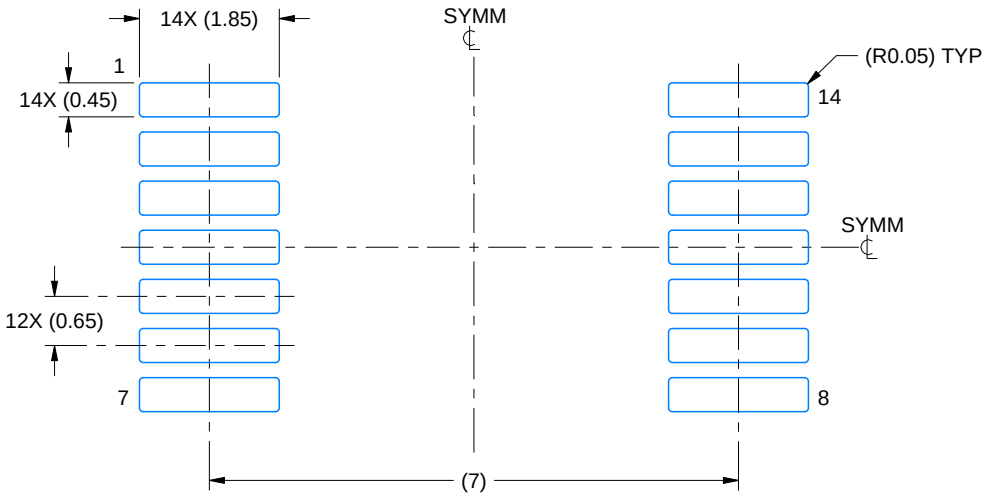
## SSOP - 2 mm max height

SMALL OUTLINE PACKAGE

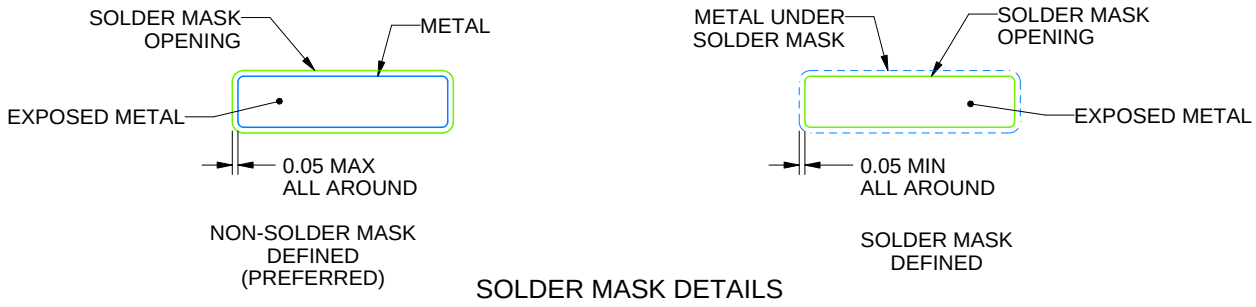



NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-150.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X

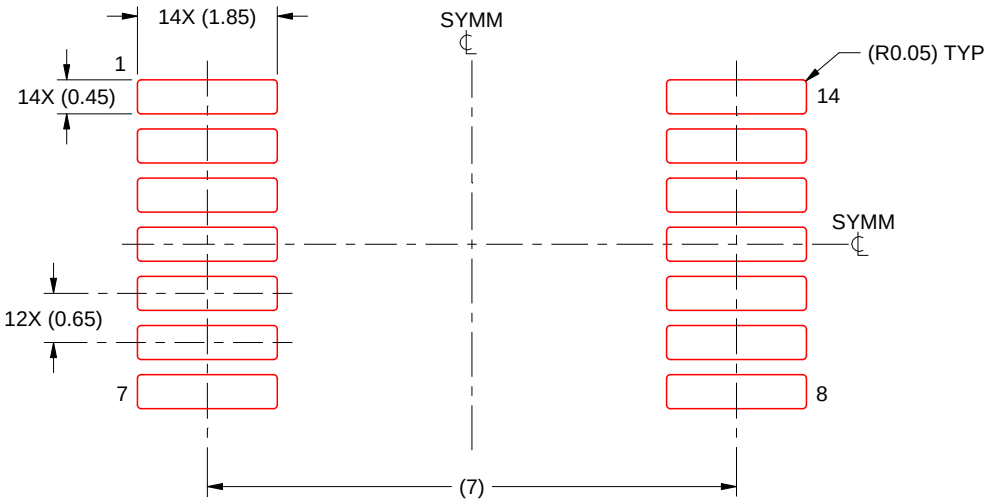



SOLDER MASK DETAILS

4220762/A 05/2024

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220762/A 05/2024

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

# GENERIC PACKAGE VIEW

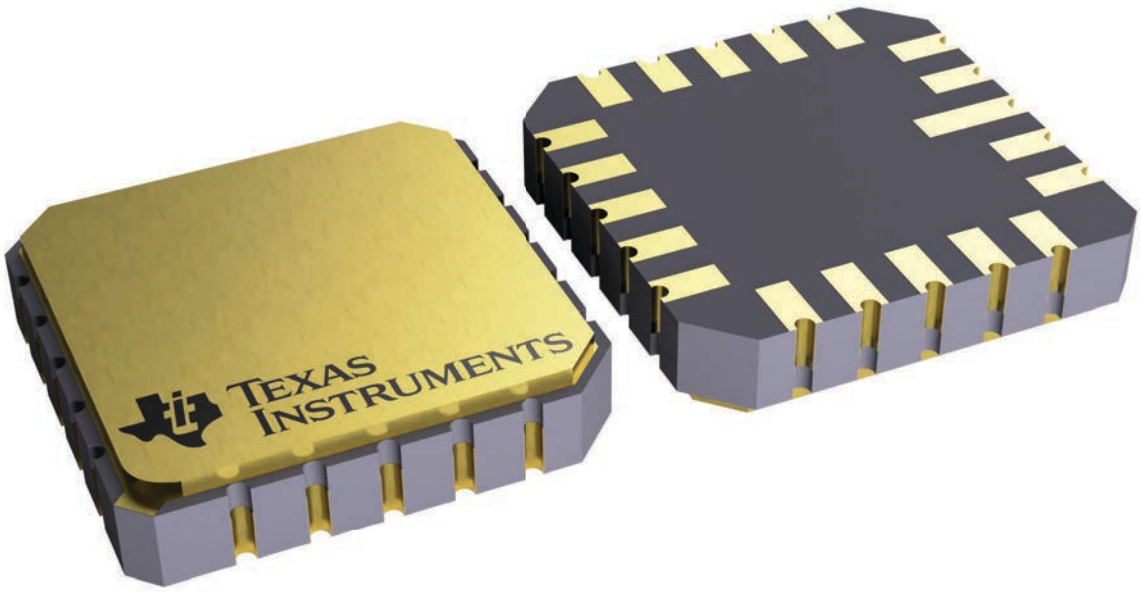
## FK 20

## LCCC - 2.03 mm max height

**8.89 x 8.89, 1.27 mm pitch**

LEADLESS CERAMIC CHIP CARRIER

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

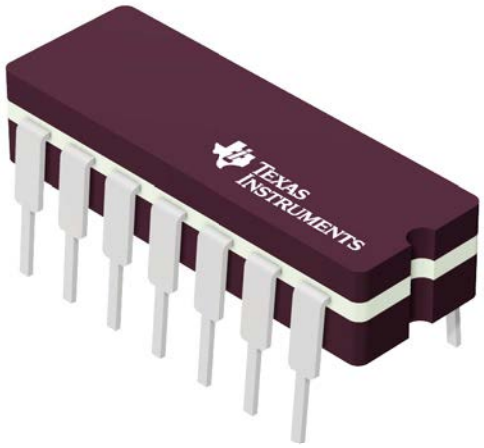



4229370VA\

# GENERIC PACKAGE VIEW

## J 14 CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE




Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

4040083-5/G

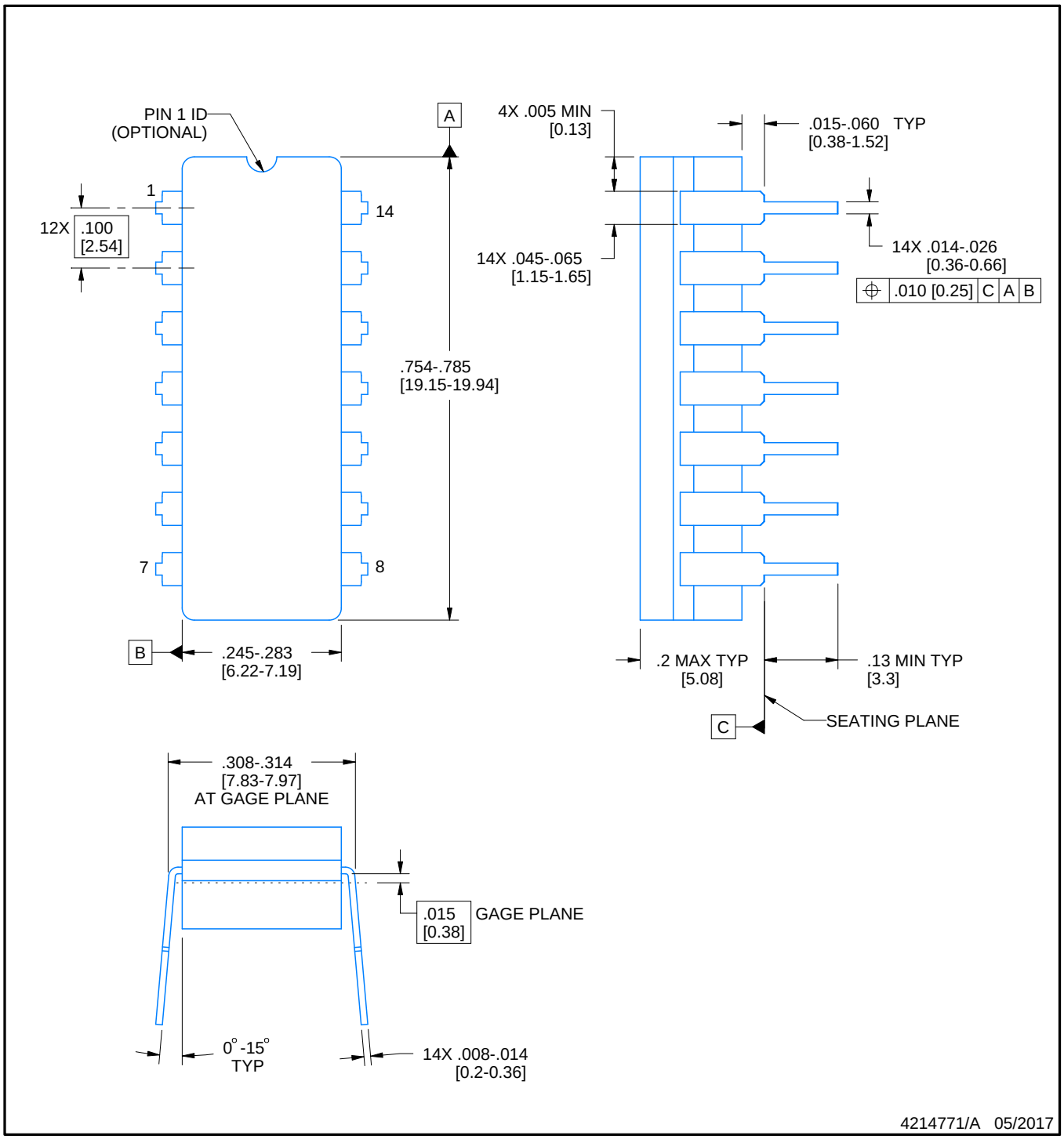


# PACKAGE OUTLINE

## J0014A

## CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE




NOTES:

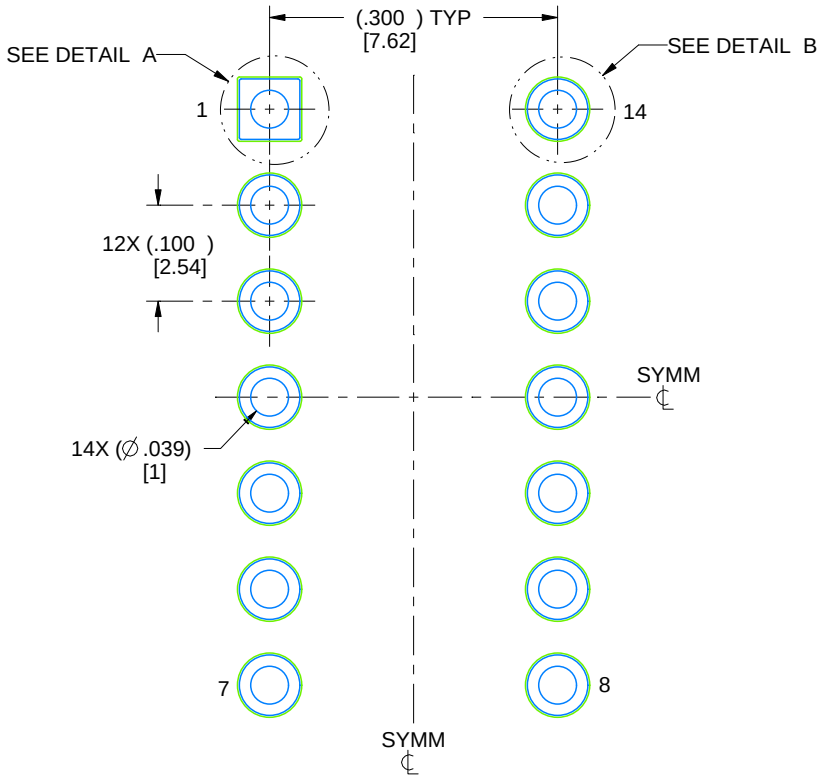
1. All controlling linear dimensions are in inches. Dimensions in brackets are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package is hermitically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification only and on press ceramic glass frit seal only.
5. Falls within MIL-STD-1835 and GDIP1-T14.

# EXAMPLE BOARD LAYOUT

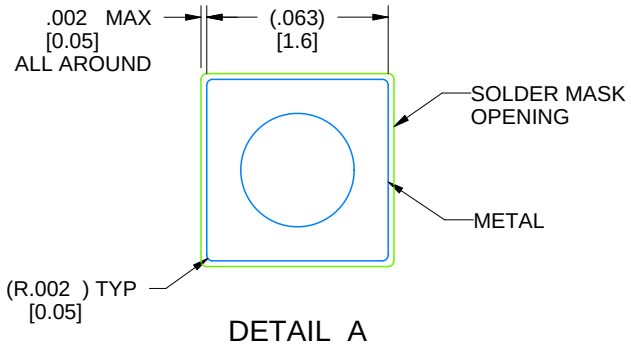
## J0014A

## CDIP - 5.08 mm max height

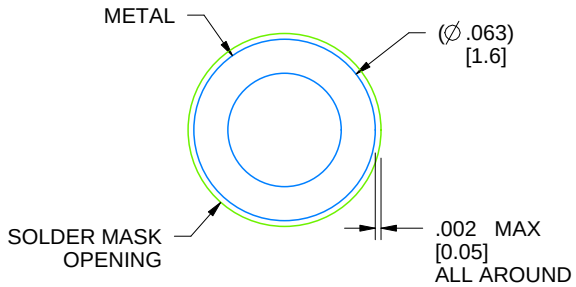
CERAMIC DUAL IN LINE PACKAGE




LAND PATTERN EXAMPLE
NON-SOLDER MASK DEFINED
SCALE: 5X




DETAIL A
SCALE: 15X



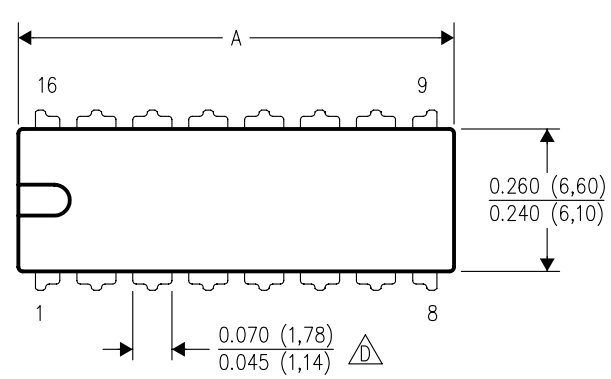

DETAIL B
13X, SCALE: 15X

4214771/A 05/2017

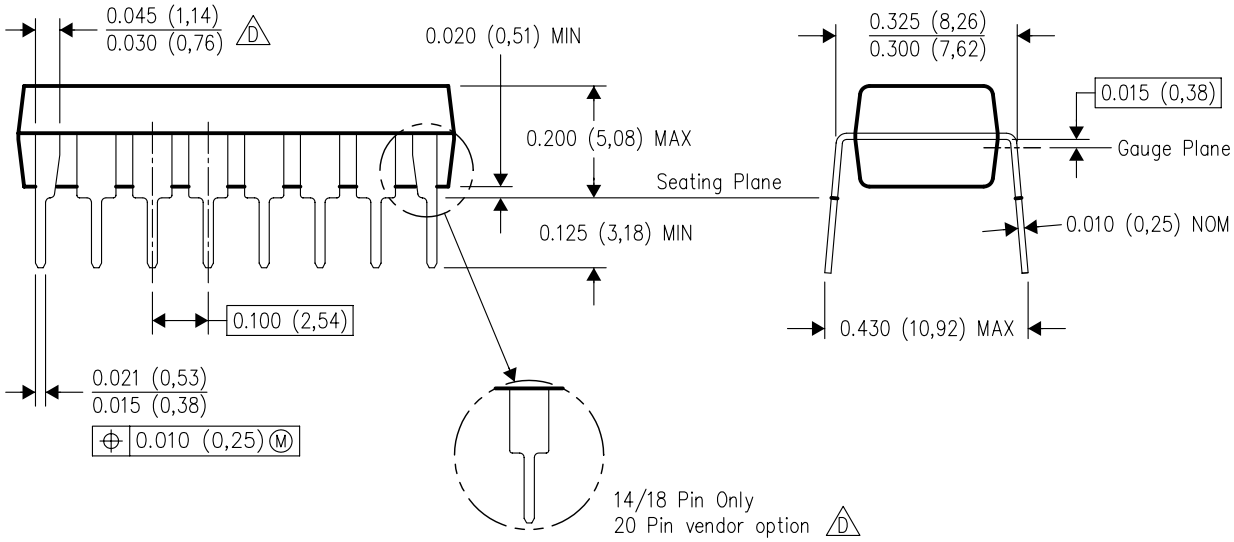
## N (R–PDIP–T**)

## PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN




| DIM \ PINS ** | 14 | 16 | 18 | 20 |
|---|---|---|---|---|
| A MAX | 0.775 (19,69) | 0.775 (19,69) | 0.920 (23,37) | 1.060 (26,92) |
| A MIN | 0.745 (18,92) | 0.745 (18,92) | 0.850 (21,59) | 0.940 (23,88) |
| MS–001 VARIATION (C) | AA | BB | AC | AD |




4040049/E 12/2002

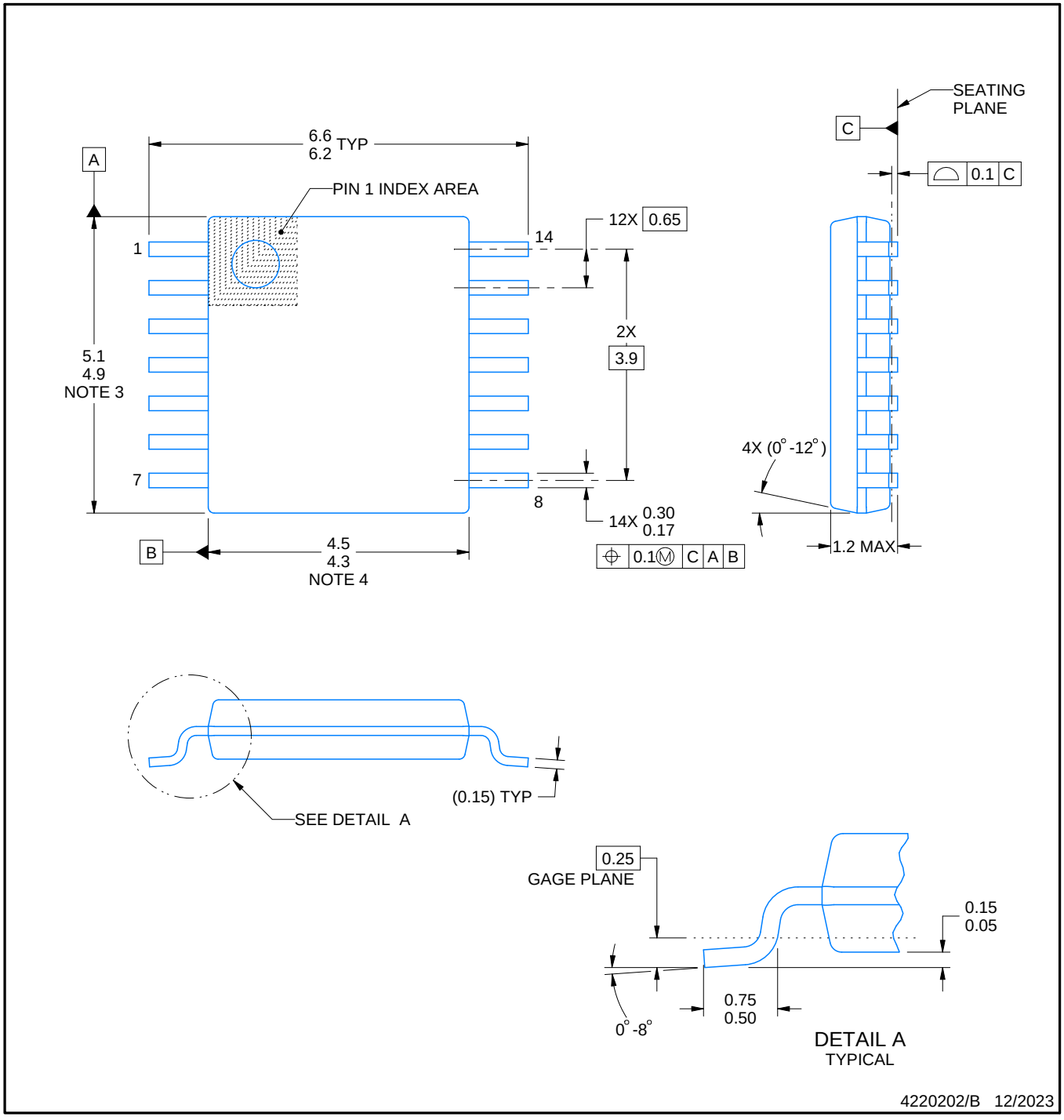
NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

# PACKAGE OUTLINE

## PW0014A

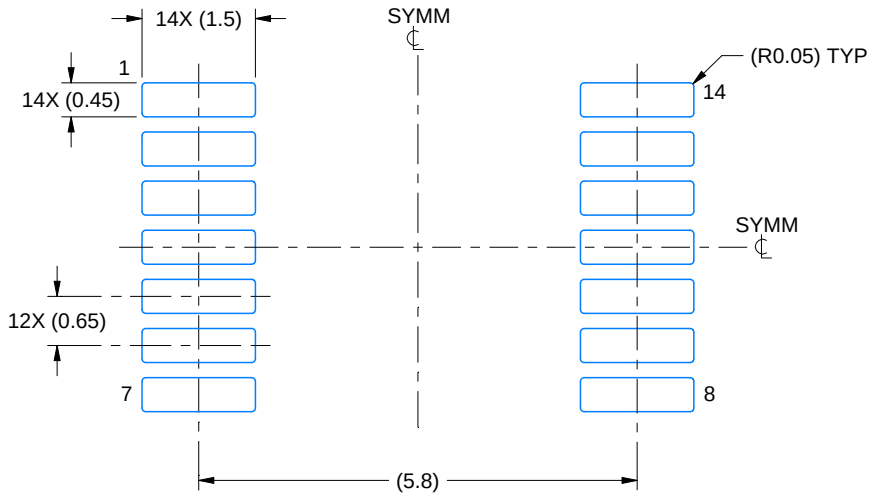
## TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE

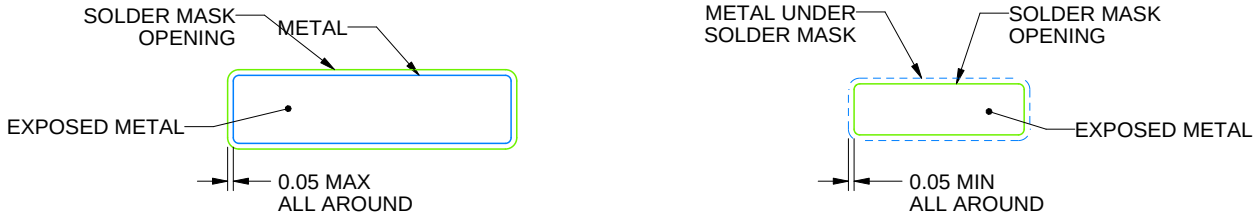



NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X

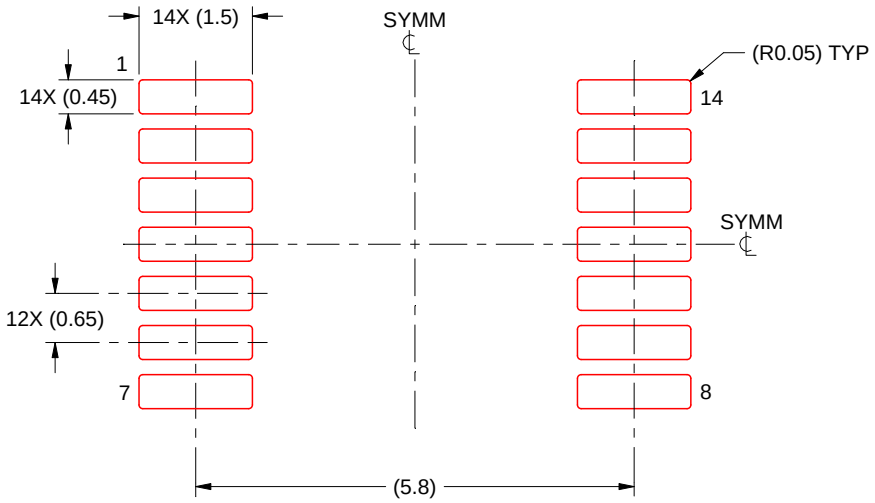



SOLDER MASK DETAILS

4220202/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220202/B 12/2023

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

# GENERIC PACKAGE VIEW

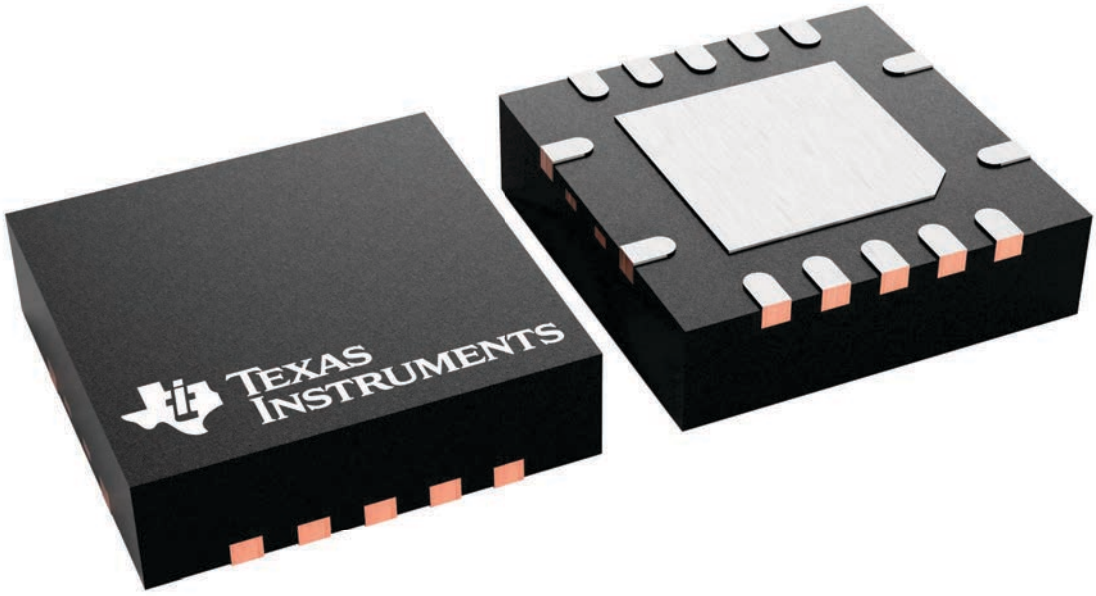
## RGY 14

## VQFN - 1 mm max height

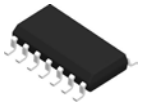
**3.5 x 3.5, 0.5 mm pitch**

PLASTIC QUAD FLATPACK - NO LEAD

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.




4231541/A

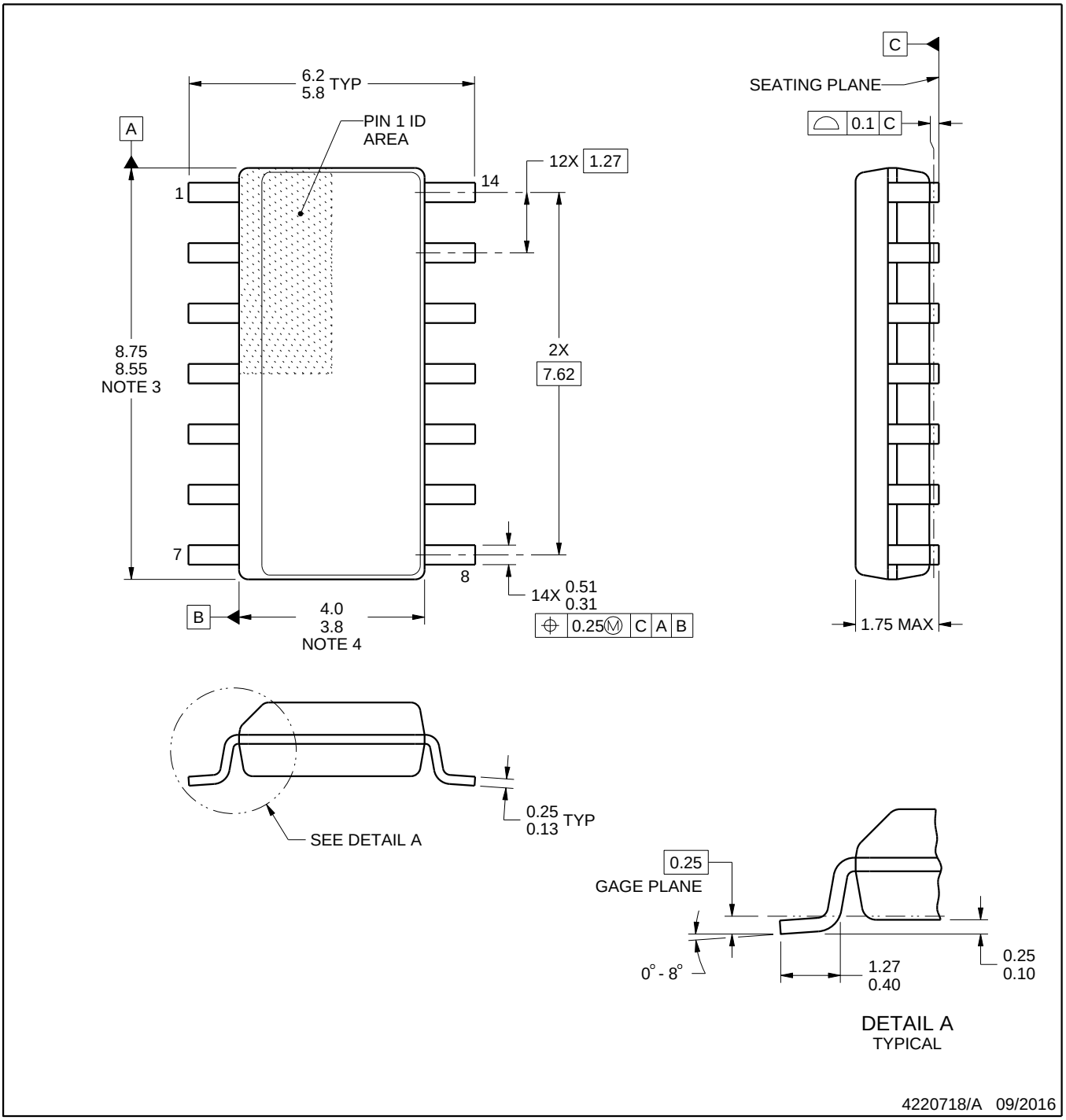


# PACKAGE OUTLINE

## D0014A

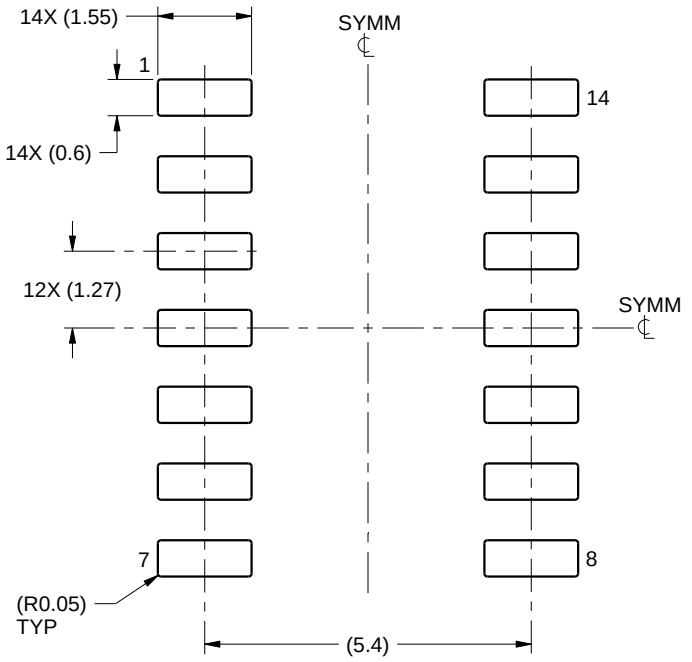
## SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT

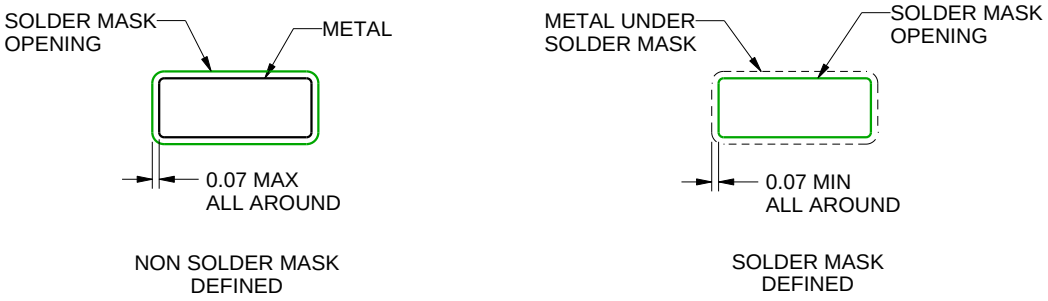



NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm, per side.
5. Reference JEDEC registration MS-012, variation AB.




LAND PATTERN EXAMPLE
SCALE:8X




SOLDER MASK DETAILS

4220718/A 09/2016

NOTES: (continued)

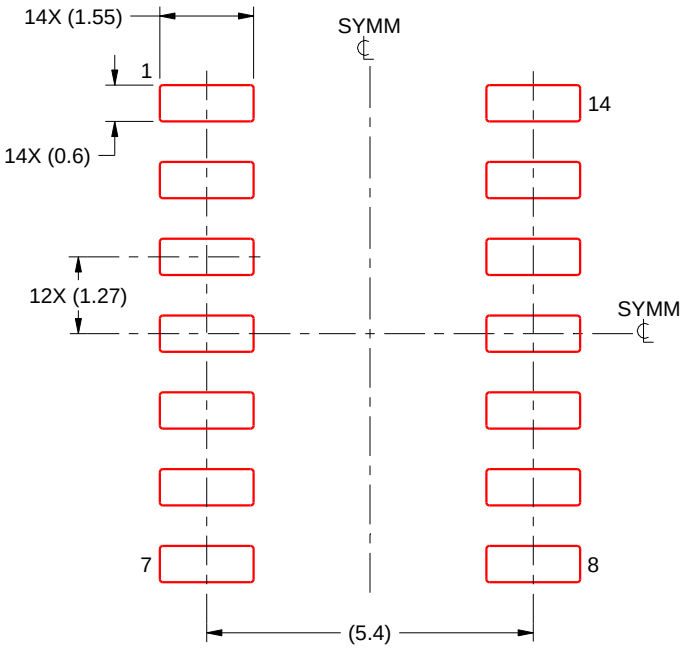
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

## D0014A

## SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:8X

4220718/A 09/2016

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

# GENERIC PACKAGE VIEW

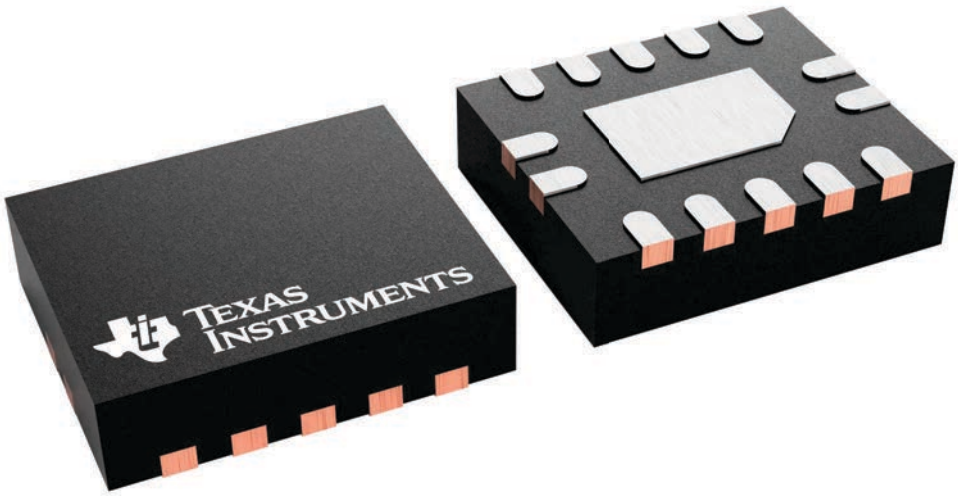
## BQA 14

## WQFN - 0.8 mm max height

**2.5 x 3, 0.5 mm pitch**

PLASTIC QUAD FLATPACK - NO LEAD

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



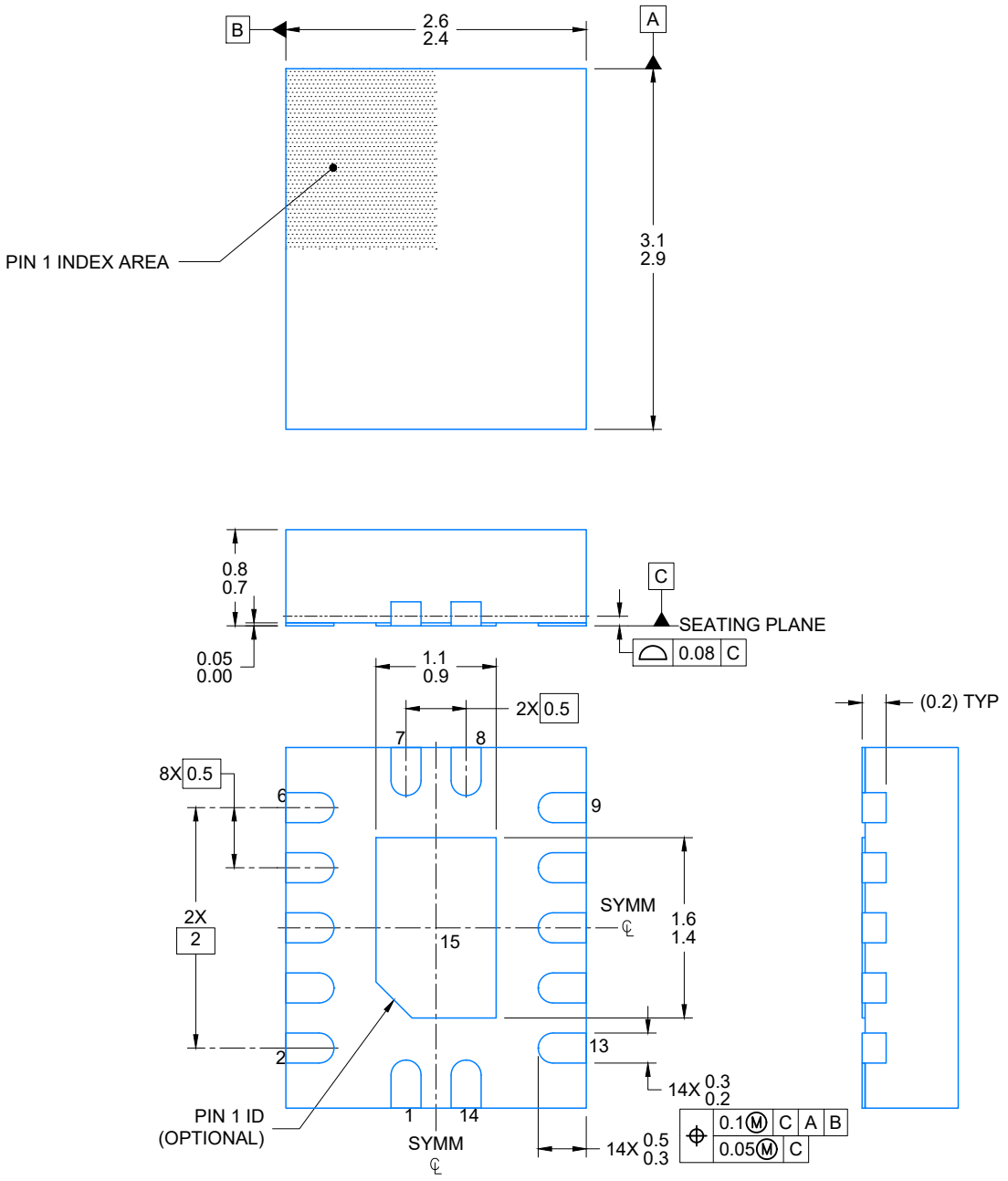

4227145/A

# PACKAGE OUTLINE

## BQA0014A

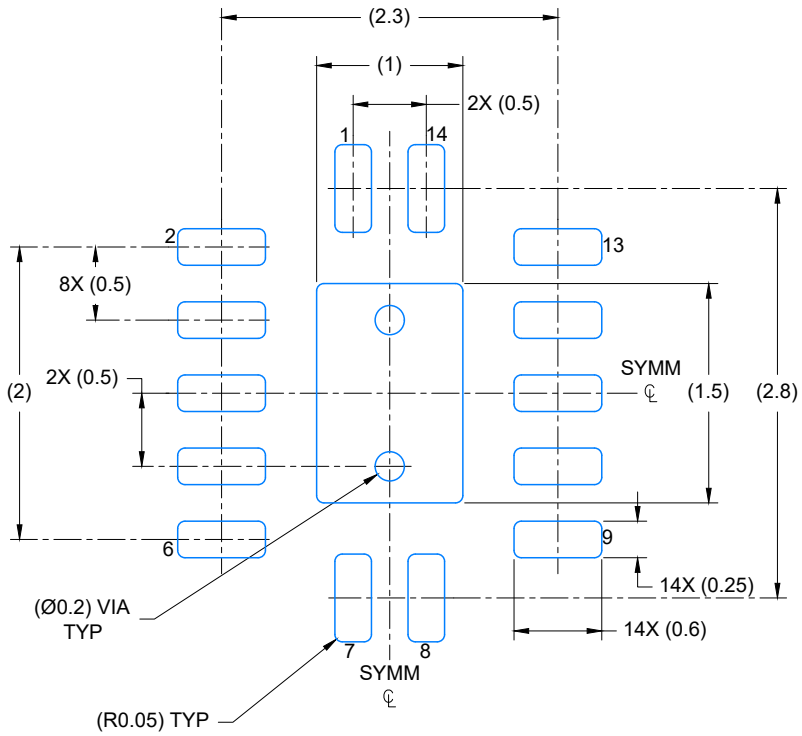
### WQFN - 0.8 mm max height

PLASTIC QUAD FLAT PACK-NO LEAD

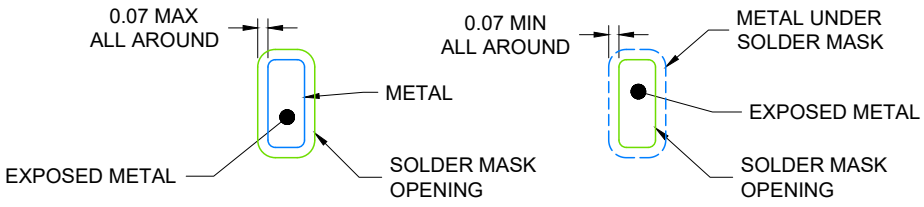



NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for optimal thermal and mechanical performance.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 20X



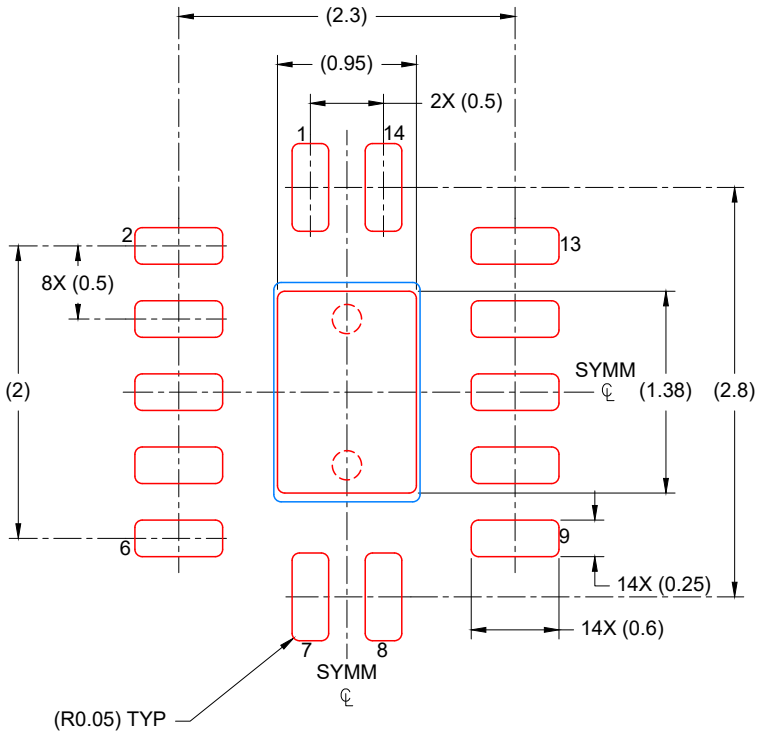

NON-SOLDER MASK
DEFINED
(PREFERRED)

SOLDER MASK
DEFINED

4224636/A 11/2018

NOTES: (continued)

4. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 (www.ti.com/lit/slua271).
5. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL

EXPOSED PAD
88% PRINTED COVERAGE BY AREA
SCALE: 20X

4224636/A 11/2018

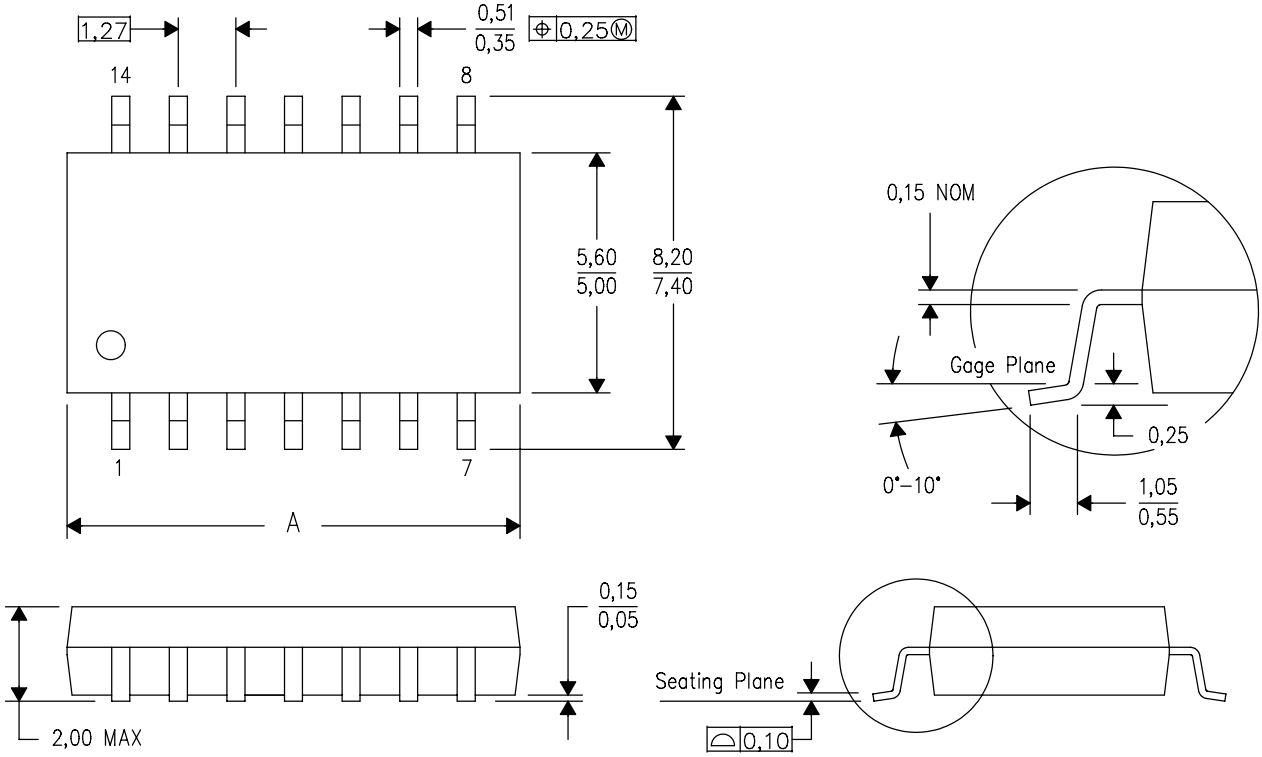
NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

## MECHANICAL DATA

**NS (R-PDSO-G**)** **PLASTIC SMALL-OUTLINE PACKAGE**

**14-PINS SHOWN**




| DIM \ PINS ** | 14 | 16 | 20 | 24 |
|---|---|---|---|---|
| A MAX | 10,50 | 10,50 | 12,90 | 15,30 |
| A MIN | 9,90 | 9,90 | 12,30 | 14,70 |

4040062/C 03/03

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

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